EXACT BOUNDED COHOMOLOGY OF MEASURE-PRESERVING HOMEOMORPHISM GROUPS OF NONORIENTABLE HYPERBOLIC MANIFOLDS

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Abstract. Let M be a closed, connected, non-orientable hyperbolic manifold. Let μ be a good measure and ω a smooth positive density on M. Denote by $\mathrm{Homeo}_0(M,\mu)$ and $\mathrm{Diff}_0(M,\omega)$ the identity components of the groups of μ-preserving homeomorphisms and ω-preserving diffeomorphisms, respectively. For every $d \geq 2$, we prove that

$$\dim \overline{EH}_b^d\big(\mathrm{Homeo}_0(M,\mu)\big) = \dim \overline{EH}_b^d\big(\mathrm{Diff}_0(M,\omega)\big) = \infty,$$

where $\overline{EH}_b^d$ denotes reduced exact bounded cohomology in degree d.

1. Introduction

Gromov's seminal work [21] established bounded cohomology as a fundamental tool in geometry and topology, with applications to the geometry of manifolds, stable commutator length, amenable groups, and symplectic geometry. In low degrees it is understood for many classes of finitely generated groups. In higher degrees, however, even the bounded cohomology of the non-abelian free group F_2 remains much less understood; see [18].

Since the early 2000s, bounded cohomology of transformation groups has received considerable attention. Degree-two exact bounded cohomology is described by homogeneous quasimorphisms modulo homomorphisms. Quasimorphisms have applications to dynamics and symplectic geometry; see [7, 23, 15]. Higher-degree bounded classes were studied by Brandenbursky-Marcinkowski [8], Kimura [24], Nitsche [28] and Brandenbursky-Menashe [9] using transfer maps from fundamental groups and braid groups. In [5], these maps were combined with external products to construct bounded cohomology classes in every degree greater than one. In this paper we apply this construction to closed non-orientable hyperbolic manifolds, both for measure-preserving homeomorphisms and for density-preserving diffeomorphisms.

Let M be a closed, connected, non-orientable hyperbolic manifold. A finite Borel measure μ on M is called *good* if it is nonatomic and positive on every nonempty open set. Let $\mathcal{T}_M = \mathrm{Homeo}_0(M,\mu)$ denote the identity component of the group of μ-preserving homeomorphisms, equipped with the compact-open topology. For a smooth positive density ω, let $\mathrm{Diff}_0(M,\omega)$ denote

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the identity component, in the C^∞ topology, of the group of ω-preserving diffeomorphisms. Throughout the paper, cohomology is taken for groups regarded as discrete. Denote by $EH_b^d(\mathcal{T}_M)$ exact bounded cohomology and by $\overline{EH}_b^d(\mathcal{T}_M)$ reduced exact bounded cohomology, with trivial real coefficients; the definitions are recalled in Section 2. Our main result is the following.

Theorem 1. *Let M be a closed, connected, non-orientable hyperbolic manifold, let μ be a good measure, and let ω be a smooth positive density on M. Then, for every $d \geq 2$,*

$$\dim \overline{EH}_b^d(\mathrm{Homeo}_0(M,\mu)) = \dim \overline{EH}_b^d(\mathrm{Diff}_0(M,\omega)) = \infty.$$

In particular, the corresponding unreduced exact bounded cohomology groups are infinite dimensional in every degree $d \geq 2$.

We divide the proof according to the dimension of M. The Oxtoby–Ulam theorem [29, Part II, Corollary 1] allows us to reduce the homeomorphism assertion to a measure induced by a smooth positive density; see Lemma 2. For the diffeomorphism assertion, we use Moser's theorem for smooth positive densities.

Let N_g denote the closed non-orientable surface of genus g, and Σ_g the closed oriented surface of genus g. In dimension two, $M = N_g$ for some $g > 2$, and its orientation double cover is Σ_{g-1}. Using this cover, we prove in Lemma 4 that, for $n \geq 3$, the degree-two exact bounded cohomology of the mapping class group of N_g with n marked points has infinite-dimensional image under restriction to the braid subgroup supported in a disc containing those points.

For the surface case of Theorem 1, we apply this lemma with $n = 3$ and choose one class with nonzero disc restriction. Pullback along the point-pushing homomorphism gives an exact degree-two bounded cohomology class on the surface braid group $B_3(N_g)$ whose restriction to the disc braid group remains nonzero. Lemma 5 shows that this class also remains nonzero on the free subgroup generated by the squares of the two standard Artin generators of the disc braid group. This restriction is detected by the local disc rotations used in the proof.

We then combine the one-point maps of [9] with the mixed transfer of [5, Section 5.1]. The transfer of the braid class supplies a fixed degree-two factor. We take cup products of its powers with the degree-two and degree-three one-point classes of Brandenbursky-Menashe [9]. The one-point factor ensures exactness, while the testing homomorphisms establish injectivity as the one-point class varies. This gives the infinite-dimensional families of reduced exact bounded cohomology classes required in the surface case.

When $\dim M \geq 3$, we choose a hyperbolically embedded free subgroup of rank two in $\pi_1(M)$ that lies in the kernel of the orientation character. We realize this subgroup by an embedded figure-eight. Since the subgroup lies

in the kernel of the orientation character, a regular neighborhood of this figure-eight is orientable. After choosing an orientation on this neighborhood, the restriction of the density is represented by a positive volume form. Moser's theorem allows us to carry out the local volume-preserving pushing construction for this form. The resulting local maps allow us to apply the transfer argument of [5, Theorem 1, case (iii)], completing the proof in higher dimensions.

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2. Preliminaries

Bounded cohomology. Throughout the paper, groups are discrete and all coefficients are trivial and real. Let K be a discrete group. We use right-homogeneous cochains, as in [5]. Thus $C_b^d(K)^K$ is the space of bounded functions $c : K^{d+1} \to \mathbb{R}$ satisfying $c(g_0h, \dots, g_dh) = c(g_0, \dots, g_d)$, $(g_0, \dots, g_d, h \in K)$. The differential is given by

$$(\delta c)(g_0, \dots, g_{d+1}) = \sum_{j=0}^{d+1} (-1)^j c(g_0, \dots, \widehat{g}_j, \dots, g_{d+1}).$$

The cohomology of this complex is $H_b^d(K)$. The supremum norm on cochains induces the canonical seminorm on $H_b^d(K)$. We write

$$\overline{H}_b^d(K) = H_b^d(K)/\{u \in H_b^d(K) : \|u\| = 0\}$$

for reduced bounded cohomology, with the induced quotient norm. For a bounded cocycle c, the notation $[c]$ denotes its class in the bounded or reduced bounded cohomology group indicated. A function $\varphi : K \to \mathbb{R}$ is a *homogeneous quasimorphism* if $\varphi(g^j) = j\varphi(g)$ for all $g \in K$ and $j \in \mathbb{Z}$, and

$$\sup_{g,h \in K} |\varphi(gh) - \varphi(g) - \varphi(h)| < \infty.$$

For such a function, $\delta\varphi$ denotes, in our convention, the bounded cocycle

$$(\delta\varphi)(g_0, g_1, g_2) = \varphi(g_0 g_1^{-1}) + \varphi(g_1 g_2^{-1}) - \varphi(g_0 g_2^{-1}).$$

Dropping the boundedness condition gives the ordinary cochain complex $C^\bullet(K)^K$, whose degree-d cohomology is $H^d(K;\mathbb{R})$. Let c_K^d denote the comparison map induced by inclusion of bounded cochains into this complex.

Following [20], let $EH_b^d(K)$ denote its kernel, called the exact bounded cohomology of K. We write $\overline{EH}_b^d(K)$ for the reduced exact bounded cohomology:

$$EH_b^d(K) = \ker\bigl(c_K^d : H_b^d(K) \longrightarrow H^d(K;\mathbb{R})\bigr),$$
$$\overline{EH}_b^d(K) = \operatorname{im}\bigl(EH_b^d(K) \longrightarrow \overline{H}_b^d(K)\bigr).$$

Thus $\overline{EH}_b^d(K)$ is $EH_b^d(K)$ modulo its zero-seminorm subspace, with the norm induced from $\overline{H}_b^d(K)$. Every class in $EH_b^2(K)$ is represented by $\delta\varphi$ for a homogeneous quasimorphism φ. For such a quasimorphism, $[\delta\varphi] = 0$ in $H_b^2(K)$ if and only if φ is a homomorphism. See [13, 19] for the basic properties.

Pullback by a group homomorphism is seminorm nonincreasing and commutes with comparison. A group isomorphism therefore induces isometric isomorphisms on reduced bounded cohomology and on reduced exact bounded cohomology.

Let $F_2 = \langle a, b\rangle$ be the free group of rank two. We shall use

$$\dim \overline{H}_b^2(F_2) = \dim \overline{H}_b^3(F_2) = \infty; \tag{1}$$

see [10, 31, 16]. Since $H^q(F_2;\mathbb{R}) = 0$ for $q \geq 2$, these are also reduced exact bounded cohomology groups.

Good measures and densities. A smooth positive density on a smooth n-manifold M is a positive smooth section of $|\Lambda^n T^*M|$. On a closed manifold it defines a good measure. On an oriented open set U, it is represented by a unique positive volume form ω_U. In particular, every Riemannian metric determines a smooth positive density.

Lemma 2. *For each M, it suffices to prove the assertion of Theorem 1 concerning* $\mathrm{Homeo}_0(M,\mu)$ *for one fixed probability measure μ induced by a smooth positive density.*

Proof. Rescaling a good measure μ does not change $\mathrm{Homeo}_0(M,\mu)$, so assume that $\mu(M) = 1$. Fix a probability measure ν induced by a smooth positive density. By the Oxtoby–Ulam theorem [29, Part II, Corollary 1], there is a homeomorphism h such that $h_*\mu = \nu$; see also [17, Theorem 3.1]. Conjugation gives a topological group isomorphism

$$h\ \mathrm{Homeo}_0(M,\mu)\, h^{-1} = \mathrm{Homeo}_0(M,\nu).$$

The pullbacks induced by this conjugation isomorphism and its inverse are inverse isometries of bounded cochain complexes. They commute with the ordinary comparison maps. Hence they induce seminorm-preserving isomorphisms on exact bounded cohomology and isometric isomorphisms on reduced exact bounded cohomology. □

For the diffeomorphism assertion, the analogous reduction follows from Moser's theorem for smooth positive densities [27, 11], applied on the orientation double cover when M is non-orientable. After rescaling to equal total mass, any two such densities are related by a diffeomorphism isotopic to the identity. Conjugation therefore identifies the corresponding density-preserving diffeomorphism groups and their identity components.

From now on we fix a hyperbolic metric on M, denote by ω_0 its normalized Riemannian density, and let μ be its induced measure. Let $p : \widetilde{M} \to M$ be the orientation double cover, with its canonical orientation, and let τ be its nontrivial deck transformation. Write $\widetilde{\omega}$ for the volume form of the lifted metric, rescaled so that $\int_{\widetilde{M}} \widetilde{\omega} = 2$. Then $\tau^*\widetilde{\omega} = -\widetilde{\omega}$, and explicitly

$$\mu(A) = \frac{1}{2} \int_{p^{-1}(A)} \widetilde{\omega} \qquad (A \subset M \text{ Borel}).$$

Thus $\mu(M) = 1$. The canonical orientation-preserving lift of every μ-preserving homeomorphism preserves the measure induced by $\widetilde{\omega}$. For $M = N_g$, this is the measure used in [9, Section 1]. All subsequent constructions and both cases of the proof of Theorem 1 use this fixed measure. In particular, $\mathrm{Diff}_0(M, \omega_0) \subset \mathrm{Homeo}_0(M, \mu)$, so the transfer classes constructed below can be restricted to this diffeomorphism subgroup. Lemma 2 then gives the result for every good measure.

External products. Let $K_1, \dots, K_k$ be groups, where $k \geq 1$, and write $p_j : K_1 \times \cdots \times K_k \to K_j$ for the coordinate projections. Denote the cup product by $\smile$. For $u_j \in H_b^{d_j}(K_j)$, define

$$u_1 \times \cdots \times u_k = p_1^* u_1 \smile \cdots \smile p_k^* u_k \in H_b^{d_1 + \cdots + d_k}(K_1 \times \cdots \times K_k).$$

Cup and external products descend to reduced bounded cohomology and satisfy

$$\|u \smile v\| \leq \|u\| \, \|v\|, \qquad \|u \times v\| \leq \|u\| \, \|v\|.$$

When all factors are a group K, the symmetric group S_k acts on K^k by

$$\sigma(g_1, \dots, g_k) = (g_{\sigma^{-1}(1)}, \dots, g_{\sigma^{-1}(k)}).$$

For $u \in H_b^d(K^k)$, put

$$\mathrm{Sym}_k(u) = \sum_{\sigma \in S_k} \sigma^* u.$$

We use this unnormalized sum also on reduced bounded cohomology.

Write $Z_2^{\ell^1}(F_2)$ for the degree-two cycles in the ℓ^1-completed bar complex, and use $\langle \cdot, \cdot \rangle$ for evaluation of bounded cohomology on these cycles. Fix a nonzero class $\theta \in H_b^2(F_2)$ and a cycle $z \in Z_2^{\ell^1}(F_2)$ such that $\langle \theta, z \rangle = 1$. Such a cycle exists by degree-two bounded-cohomology–ℓ^1-homology duality; see [5, Lemmas 7–8] and [26].

For $m \geq 0$ and $q \in \{2,3\}$, put $d = 2m+q$ and $k = m+1$. The notation $\theta^{\times m}$ means the external product of m copies of θ; when $m = 0$, this is the degree-zero unit. Define

$$E_m^q : H_b^q(F_2) \longrightarrow H_b^d(F_2^k), \qquad E_m^q(v) = \theta^{\times m} \times v, \tag{2}$$

and

$$T_m^q : H_b^q(F_2) \longrightarrow H_b^d(F_2^k), \qquad T_m^q(v) = \mathrm{Sym}_k(\theta^{\times m} \times v). \tag{3}$$

Both maps descend to reduced bounded cohomology. We use the same notation for these induced maps.

Proposition 3 ([5, Lemma 8 and Proposition 9])**.** *For $q = 2, 3$ and every $m \geq 0$, the maps E_m^q and T_m^q are injective. Their reduced versions have bounded linear left inverses.*

Proof. We recall the calculation of the left inverses from [5, Lemma 8 and proof of Proposition 9]. When $m = 0$, both maps are the identity. For $m \geq 1$, let $Z_m = z^{\times m} \in Z_{2m}^{\ell^1}(F_2^m)$ be the chain cross product of m copies of z. The bounded slant-product map

$$L_{Z_m}^* : H_b^{2m+q}(F_2^m \times F_2) \longrightarrow H_b^q(F_2)$$

evaluates the first m factors against Z_m; see [5, Lemma 8]. It satisfies

$$L_{Z_m}^* E_m^q(v) = v.$$

For the symmetric product, the calculation in [5, Proposition 9] gives

$$L_{Z_m}^* T_m^3(v) = m!v, \qquad L_{Z_m}^* T_m^2(v) = m!\big(v + m\,\langle v, z\rangle\,\theta\big).$$

The operator $P : H_b^2(F_2) \to H_b^2(F_2)$ given by $P(v) = \langle v, z\rangle\,\theta$ satisfies $P^2 = P$, since $\langle \theta, z\rangle = 1$. Thus

$$(\mathrm{id} + mP)^{-1} = \mathrm{id} - \frac{m}{m+1}P.$$

Thus $L_{Z_m}^*$ is a left inverse of E_m^q, $(1/m!)L_{Z_m}^*$ is a left inverse of T_m^3, and $(1/m!)(\mathrm{id} - \frac{m}{m+1}P)L_{Z_m}^*$ is a left inverse of T_m^2. These identities hold before reduction, and all three left inverses descend to bounded maps on reduced bounded cohomology. □

The product transfer. Fix $x_0 \in M$ and put

$$\pi = \pi_1(M, x_0), \qquad G = \pi_M = \pi / Z(\pi).$$

Here $Z(\pi)$ is the center of π. For the manifolds considered here $Z(\pi) = 1$, so $G = \pi$. Here $Z(\pi)$ is the center of π. For the manifolds considered here $Z(\pi) = 1$, so $G = \pi$. Fix an auxiliary Riemannian metric on M. Outside the cut locus of x_0, let α_x be the unique minimizing geodesic from x_0 to x. The cut locus has μ-measure zero, since μ is induced by a smooth positive density. Choose the remaining paths measurably.

For $f \in \mathcal{T}_M$, choose an isotopy f_t from the identity to f. The trajectory cocycle is

$$\gamma : \mathcal{T}_M \times M \longrightarrow G, \qquad \gamma(f,x) = [\alpha_x * \{f_t(x)\} * \alpha_{f(x)}^{-1}]^{-1}.$$

The bracket denotes the based homotopy class in G, and $*$ denotes path concatenation from left to right, with multiplication $[\alpha][\beta] = [\alpha * \beta]$. This is the construction of [9, Section 3], with the inverse-loop convention of [5, equation (22)]. With multiplication $fg = f \circ g$, it satisfies

$$\gamma(fg,x) = \gamma(f,gx)\gamma(g,x).$$

The evaluation of a loop of homeomorphisms lies in $Z(\pi)$. Thus the cocycle is independent of the chosen isotopies.

For $k \geq 1$, equip M^k with the product measure μ^k and the diagonal action on $\mathbf{x} = (x_1, \dots, x_k)$ given by $h\mathbf{x} = (h(x_1), \dots, h(x_k))$. Define

$$\gamma_k : \mathcal{T}_M \times M^k \longrightarrow G^k, \qquad \gamma_k(f,\mathbf{x}) = (\gamma(f,x_1), \dots, \gamma(f,x_k)).$$

Define the cochain map $\Gamma'_{b,k} : C_b^d(G^k)^{G^k} \to C_b^d(\mathcal{T}_M)^{\mathcal{T}_M}$ by

$$(\Gamma'_{b,k}c)(f_0, \dots, f_d) = \int_{M^k} c\big(\gamma_k(f_0,\mathbf{x}), \dots, \gamma_k(f_d,\mathbf{x})\big)\, \mathrm{d}\mu^k(\mathbf{x}). \tag{4}$$

Right homogeneity follows from the cocycle identity and the change of variables $\mathbf{x} \mapsto h\mathbf{x}$. Integration commutes with δ, and $\left\|\Gamma'_{b,k}c\right\|_\infty \leq \|c\|_\infty$. We denote the induced maps by

$$\Gamma_{b,k}^d : H_b^d(G^k) \longrightarrow H_b^d(\mathcal{T}_M), \qquad \overline{\Gamma}_{b,k}^d : \overline{H}_b^d(G^k) \longrightarrow \overline{H}_b^d(\mathcal{T}_M).$$

By [8, Lemma 2.2], for each $f \in \mathcal{T}_M$ there is a finite set $S_f \subset G$ such that

$$\gamma(f,x) \in S_f \qquad \text{for almost every } x \in M.$$

The proof applies without an orientability assumption. We use this finite-image property as in [5, proof of Lemma 10, equation (24)]. For each fixed tuple $f_0, \dots, f_d \in \mathcal{T}_M$, the map

$$\mathbf{x} \longmapsto \big(\gamma_k(f_0,\mathbf{x}), \dots, \gamma_k(f_d,\mathbf{x})\big)$$

takes values almost everywhere in the finite set $S_{f_0}^k \times \cdots \times S_{f_d}^k$. Hence, even for an ordinary cochain c which is not bounded, the integrand in (4) takes only finitely many values outside a null set. Since μ is finite, this integrand is integrable. Thus (4) also defines a map on ordinary cochains. Denote the resulting maps by

$$\Gamma'_k : C^d(G^k)^{G^k} \longrightarrow C^d(\mathcal{T}_M)^{\mathcal{T}_M}, \qquad \Gamma_k^d : H^d(G^k;\mathbb{R}) \longrightarrow H^d(\mathcal{T}_M;\mathbb{R}).$$

The comparison identity

$$c_{\mathcal{T}_M}^d \Gamma_{b,k}^d = \Gamma_k^d c_{G^k}^d$$

shows that the transfer preserves exact classes; see [8, Lemma 2.2] and [5, Lemma 10]. These constructions do not require an orientation of M.

Configuration spaces. For $g > 2$ and $r \geq 1$, let

$$X_r(N_g) = \{(x_1, \dots, x_r) \in N_g^r : x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group S_r acts by permuting the coordinates. With chosen basepoints write

$$P_r(N_g) = \pi_1(X_r(N_g)), \qquad B_r(N_g) = \pi_1(X_r(N_g)/S_r)$$

for the pure and full braid groups, respectively. We use the same notation for configuration spaces and braid groups of a disc, using its interior when the disc is closed. We put

$$\nu = \mu^3|_{X_3(N_g)}, \qquad \mathcal{T}_{N_g} = \mathrm{Homeo}_0(N_g, \mu).$$

Since μ is nonatomic and $\mu(N_g) = 1$, the measure ν also has total mass one. Choose measurable closing paths in $X_3(N_g)$ from a fixed base configuration. Applying an isotopy to all three points and closing the resulting path, with the same inverse-loop convention as for γ, defines the trajectory cocycle

$$\gamma_{N_g,3} : \mathcal{T}_{N_g} \times X_3(N_g) \longrightarrow P_3(N_g) \subset B_3(N_g).$$

The pure braid groups $P_r(N_g)$ are center-free for every $r \geq 1$; see [30, proof of Proposition 1.6]. Thus $\gamma_{N_g,3}$ is independent of the chosen isotopy.

We define the cochain map

$$\Gamma'_{N_g,3} : C_b^d(B_3(N_g))^{B_3(N_g)} \longrightarrow C_b^d(\mathcal{T}_{N_g})^{\mathcal{T}_{N_g}}$$

by (4), replacing M^k, μ^k, γ_k by $X_3(N_g), \nu, \gamma_{N_g,3}$, respectively; see [24]. Following [5, Sections 4-5], we use the notation

$$\Gamma^d_{N_g,3} : \overline{H}_b^d(B_3(N_g)) \longrightarrow \overline{H}_b^d(\mathcal{T}_{N_g})$$

for the induced reduced transfer. For $m \geq 0$, the product space $X_3(N_g)^m \times N_g$, with measure $\nu^m \times \mu$, carries the diagonal action of $\mathcal{T}_{N_g}$. The product of the three-point cocycles on the first m factors and the one-point cocycle on the last factor takes values in $B_3(N_g)^m \times \pi_1(N_g)$. Integrating bounded cochains against this product cocycle defines the mixed transfer used in Section 3. All cochains integrated in these configuration-space constructions are bounded.

3. Proof of Theorem 1 in dimension two

Let $g > 2$, $G = \pi_1(N_g)$ and $\overline{\Gamma}^q_{b,1}$ denotes the one-point transfer on N_g.

The one-point classes. We briefly recall the construction of Brandenbursky–Menashe [9] and the proof of their main theorem, which give the degree-two and degree-three classes used below. Choose an oriented one-holed torus $Q \cong \Sigma_{1,1} \subset N_g$ such that $N_g \setminus \operatorname{int} Q \cong N_{g-2,1}$. Here $N_{g-2,1}$ is the compact non-orientable surface of genus $g-2$ with one boundary component. Choose a common basepoint for the meridian and longitude of Q and a path from x_0 to it. Let $i : F_2 \hookrightarrow \pi_1(N_g)$ send a, b to their based homotopy classes. Then

$$\pi_1(N_g) = F_2 *_{\mathbb{Z}} \pi_1(N_{g-2,1}).$$

For $q = 2, 3$, the isometric section theorem of [12] gives isometric linear sections of restriction

$$J_q : \overline{H}^q_b(F_2) \longrightarrow \overline{H}^q_b(\pi_1(N_g)), \qquad i^* J_q = \mathrm{id};$$

see [9, Proposition 4.10]. Since N_g is closed, aspherical and non-orientable,

$$H^q(\pi_1(N_g); \mathbb{R}) = 0 \qquad (q \geq 2). \tag{5}$$

Thus $J_q(v) \in \overline{EH}^q_b(\pi_1(N_g))$ for every $v \in \overline{H}^q_b(F_2)$. Put

$$\Theta_q : \overline{H}^q_b(F_2) \longrightarrow \overline{EH}^q_b(\mathcal{T}_{N_g}), \qquad \Theta_q(v) = \overline{\Gamma}^q_{b,1}(J_q(v)).$$

For sufficiently small $\delta > 0$, let $s_\delta : F_2 \to \mathcal{T}_{N_g}$ be the annular pushing homomorphisms of [9, Lemma 4.12]. Their defining isotopies are supported in annuli about the chosen meridian and longitude of Q, with transition collars whose measure tends to zero as $\delta \to 0$. Choose the pushing directions so that, for almost every x in the overlap of the two annuli outside their transition collars, there is an element $u_x \in \pi_1(N_g)$ such that

$$\gamma(s_\delta(w), x) = u_x \, i(w) \, u_x^{-1} \qquad \text{for every } w \in F_2.$$

On Q, the density defining μ is a positive area form, so the boundary version of Moser's theorem [27, 2] gives this local construction. There is a constant $\kappa > 0$ such that, for $q = 2, 3$ and every fixed $v \in \overline{H}^q_b(F_2)$,

$$\| s_\delta^* \Theta_q(v) - \kappa v \| \longrightarrow 0 \qquad (\delta \to 0). \tag{6}$$

The maps and the constant are independent of q and v. If $\Theta_q(v) = 0$, then (6) gives $\|\kappa v\| = 0$. Since we work in reduced bounded cohomology and $\kappa > 0$, this implies $v = 0$, and thus Θ_q is injective.

Restriction to disc braid groups. Let $\mathrm{Mod}(N_g, n)$ denote the full mapping class group of N_g with n marked points. Fix an embedded oriented disc $D \subset N_g$ whose interior contains these points. Extension by the identity gives an injective homomorphism

$$\iota_D : B_n(D) \hookrightarrow \mathrm{Mod}(N_g, n).$$

Indeed, this map is the composition of the homomorphism $B_n(D) \to B_n(N_g)$ induced by the inclusion $D \hookrightarrow N_g$, which is injective by [30, Theorem 2.3], and the point-pushing homomorphism $B_n(N_g) \to \mathrm{Mod}(N_g, n)$, which is injective for $g > 2$ by [25, Theorem 1.1, diagram (1.4)].

Lemma 4. *Let $g > 2$ and $n \geq 3$. Then the image of the restriction map*

$$\iota_D^* : EH^2_b(\mathrm{Mod}(N_g, n)) \longrightarrow EH^2_b(B_n(D))$$

is infinite dimensional. The same assertion holds for reduced exact bounded cohomology.

Proof. Choose a smaller disc $E \subset D$ containing exactly three marked points. Regard $B_3(E)$ as a subgroup of $B_n(D)$ by fixing the remaining strands. Let

σ_1, σ_2 be its standard Artin generators, and let Δ_3^2 denote the full twist. For every integer $r \geq 3$, put

$$\Delta_3^2 = (\sigma_1\sigma_2)^3, \qquad u_r = \sigma_2^{-1}\sigma_1\sigma_2^{-r}\sigma_1^2, \qquad v_r = u_r^6(\Delta_3^2)^{r-2}. \tag{7}$$

The exponent-sum homomorphism $B_3(E) \to \mathbb{Z}$, which sends each Artin generator to 1, is the abelianization. The exponent sums of u_r and Δ_3^2 are $2-r$ and 6, respectively. Hence the exponent sum of v_r is zero, and therefore

$$v_r \in [B_3(E), B_3(E)] \subset [B_n(D), B_n(D)]. \tag{8}$$

Put

$$R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

and write $[A]$ for the projective class of a matrix A. Use the identification $B_3(E)/\langle\Delta_3^2\rangle \cong \mathrm{PSL}(2,\mathbb{Z})$ given by $\sigma_1 \mapsto [R]$ and $\sigma_2 \mapsto [L^{-1}]$. Then u_r maps to $[M_r]$, where

$$M_r = LRL^rR^2 = \begin{pmatrix} r+1 & 2r+3 \\ 2r+1 & 4r+4 \end{pmatrix}.$$

Its trace is $5(r+1) > 2$, so v_r is pseudo-Anosov on the three-punctured disc E, with projective matrix $[M_r]^6$. Twists about ∂E are powers of Δ_3^2 and therefore have trivial image in this quotient. A conjugacy by a homeomorphism of the marked disc, possibly reversing orientation, induces a conjugacy of projective matrices in $\mathrm{PGL}(2,\mathbb{Z})$.

We first establish two properties of the projective matrices $[M_r]$ in $\mathrm{PGL}(2,\mathbb{Z})$: no nonzero power of $[M_r]$ is conjugate to its inverse, and, if $r \neq s$, no nonzero power of $[M_r]$ is conjugate to a nonzero power of $[M_s]$.

Decompose the cyclic word into maximal blocks of consecutive identical letters. To each block L^a assign the integer $-a$, and to each block R^b assign the integer b, keeping the order of the blocks. For $M_r = LRL^rR^2$, the resulting sequence is $(-1, 1, -r, 2)$. By the normal form of [1, Section 3.1.1, Propositions 3–4], conjugacy to the inverse in $\mathrm{PGL}(2,\mathbb{Z})$ requires this cyclic sequence to agree with its reversal or its negative reversal. Neither is possible. The negative block of length r is preceded by a positive block of length 1 and followed by one of length 2, whereas reversal interchanges these neighbors. Negative reversal has a positive block of length r, which the original sequence lacks. The same argument applies to every repetition of the sequence. Thus no nonzero power of $[M_r]$ is conjugate to its inverse.

The largest absolute block length is r. It is unchanged by repetition, cyclic permutation, reversal, or change of all signs. Consequently, if $r \neq s$, no nonzero powers of $[M_r]$ and $[M_s]$ are conjugate in $\mathrm{PGL}(2,\mathbb{Z})$.

Let $p : \Sigma_{g-1} \to N_g$ be the orientation cover. Its nontrivial deck transformation reverses orientation and will be called the deck reflection. Write $\mathrm{Mod}^+(\Sigma_{g-1}, 2n)$ for the group of orientation-preserving mapping classes of

the cover with the $2n$ lifted marked points, which may be permuted. The canonical orientation-preserving lift gives an injective homomorphism

$$\Lambda : \mathrm{Mod}(N_g, n) \hookrightarrow \mathrm{Mod}^+(\Sigma_{g-1}, 2n);$$

see [25, Theorem 1.1].

A suitable positive power of a mapping class admits a Nielsen–Thurston decomposition as a commuting product of pseudo-Anosov mapping classes supported on subsurfaces and powers of Dehn twists; we call these factors components. See [3, Section 4]. For the components below, conjugacy is taken in $\mathrm{Mod}^+(\Sigma_{g-1}, 2n)$. Following the description preceding [6, Lemma 4.11], a component is *chiral* if it is not conjugate to its inverse. We will verify that every positive power of each pseudo-Anosov component below is chiral. By [6, Lemma 4.11], a mapping class is detected by a homogeneous quasimorphism if the Nielsen–Thurston decomposition of a suitable positive power is nonempty, all its components are chiral, and no nonzero powers of distinct components are conjugate.

Let $f_r = \Lambda(\iota_D(v_r))$ and E^+, E^- be the two lifts of E, labeled so that $p|_{E^+} : E^+ \to E$ is orientation-preserving. The mapping class f_r preserves each of these discs and is the identity on their complement. Its Nielsen–Thurston decomposition has two pseudo-Anosov components, denoted by h_r^+, h_r^-, supported on E^+, E^-, respectively, together with non-zero powers of Dehn twists about their boundaries. Let τ denote the deck reflection. Since f_r commutes with τ and $\tau(E^+) = E^-$, we choose the pseudo-Anosov components so that $h_r^- = \tau h_r^+ \tau^{-1}$. Thus their restrictions to E^+ and E^- are conjugate by the orientation-reversing homeomorphism $\tau|_{E^+} : E^+ \to E^-$. For a pseudo-Anosov component h, let $\lambda(h) > 1$ denote the stretch factor of its restriction to its supporting disc. Thus the two components have a common stretch factor

$$\lambda(h_r^+) = \lambda(h_r^-) = \lambda_r > 1.$$

A conjugacy between nonzero powers of any two members of the family $\{h_r^+, h_r^- : r \geq 3\}$ carries the boundary of one supporting disc to the boundary of the other, up to isotopy. Indeed, these boundaries are their canonical reduction curves, which are preserved by conjugacy; see [3, Section 4]. The disc side cannot be exchanged with the complementary surface of positive genus. Thus, after an isotopy, the conjugating homeomorphism restricts to a homeomorphism between the supporting discs. In particular, when the supporting disc is the same on both sides, the conjugacy of the restrictions is induced by a homeomorphism of that disc.

Under the orientation-preserving identification $E^+ \to E$, the restriction of f_r is v_r. Passing to the component h_r^+ removes only boundary twists, which are trivial in $B_3(E)/\langle \Delta_3^2 \rangle$. Since

$$v_r = u_r^6 (\Delta_3^2)^{r-2} \longmapsto [M_r]^6,$$

the restriction of h_r^+ has projective matrix $[M_r]^6$. The deck reflection conjugates the restriction of h_r^+ to that of h_r^-. If $(h_r^+)^a$ were conjugate to $(h_r^+)^{-a}$ for some $a \geq 1$, restricting the conjugacy to E^+ would make $[M_r]^{6a}$ conjugate to its inverse in $\mathrm{PGL}(2,\mathbb{Z})$, contrary to the matrix argument above. The same argument on E^-, identified with E^+ by the deck reflection, applies to h_r^-. Thus every positive power of each $h_r^\pm$ is chiral.

Suppose that $(h_r^+)^a$ is conjugate to $(h_r^-)^{-b}$, where $a, b \geq 1$. Stretch factors are invariant under conjugacy, and

$$\lambda(h^j) = \lambda(h)^{|j|} \qquad (j \neq 0).$$

Consequently,

$$\lambda_r^a = \lambda\big((h_r^+)^a\big) = \lambda\big((h_r^-)^{-b}\big) = \lambda_r^b.$$

Since $\log \lambda_r > 0$, we obtain $a = b$. Let H be an orientation-preserving homeomorphism representing this conjugacy, so that

$$H(h_r^+)^a H^{-1} = (h_r^-)^{-a}.$$

Here and below, the equalities are equalities of mapping classes. As explained above, after an isotopy we may assume that $H(E^+) = E^-$. The deck reflection τ maps E^- back to E^+. Consequently, τH preserves E^+. Our choice of the components gives $\tau h_r^- \tau^{-1} = h_r^+$. Therefore

$$(\tau H)(h_r^+)^a(\tau H)^{-1} = \tau\big(H(h_r^+)^a H^{-1}\big)\tau^{-1} = \tau(h_r^-)^{-a}\tau^{-1} = (h_r^+)^{-a}.$$

Thus the restriction of τH to E^+ would conjugate the restriction of $(h_r^+)^a$ to its inverse.

We now apply the projective matrix representation. Under the identification $E^+ \to E$, the restriction of h_r^+ has matrix $[M_r]^6$, so its a-th power has matrix $[M_r]^{6a}$. The projective matrix representation extends to orientation-reversing homeomorphisms of the three-punctured disc, with values in $\mathrm{PGL}(2,\mathbb{Z})$. Since τH reverses orientation, let $C \in \mathrm{PGL}(2,\mathbb{Z})$ denote the projective matrix of its restriction to E^+. The preceding conjugacy would then give

$$C[M_r]^{6a}C^{-1} = [M_r]^{-6a}.$$

This contradicts the fact proved above that no nonzero power of $[M_r]$ is conjugate to its inverse in $\mathrm{PGL}(2,\mathbb{Z})$. Thus no positive power of one component is conjugate to a negative power of the other.

Fix $r \geq 3$. The construction in the proof of [3, Theorem 4.2], together with [3, Proposition 4.4], gives a homogeneous quasimorphism φ on $\mathrm{Mod}^+(\Sigma_{g-1}, 2n)$ such that $\varphi(h_r^+) \neq 0$ and φ vanishes on every boundary-twist factor in the decomposition of f_r. If h_r^+ and h_r^- have no conjugate nonzero powers, we may also choose $\varphi(h_r^-) = 0$. If nonzero powers of h_r^+ and h_r^- are conjugate, the preceding argument shows that their exponents have the same sign. Equality of stretch factors then shows that their exponents have the same absolute value. Thus $(h_r^+)^a$ and $(h_r^-)^a$ are conjugate for some $a \geq 1$, and homogeneity

and conjugacy invariance give $\varphi(h_r^+) = \varphi(h_r^-) \neq 0$. Since homogeneous quasimorphisms are additive on commuting elements, in both cases

$$\varphi(f_r) = \varphi(h_r^+) + \varphi(h_r^-) \neq 0.$$

Similarly, a conjugacy between nonzero powers of pseudo-Anosov components belonging to r and s restricts to a homeomorphism between their supporting discs. Use the covering maps to identify both discs with E. The resulting homeomorphism of E may reverse orientation, but it would give a conjugacy in $\mathrm{PGL}(2,\mathbb{Z})$ between nonzero powers of $[M_r]$ and $[M_s]$. When $r \neq s$, this contradicts the different maximal block lengths established above.

Fix $N \geq 3$. Apply [3, Proposition 4.4] to the finite collection of components of $f_3, \ldots, f_N$. For each i with $3 \leq i \leq N$, we obtain a homogeneous quasimorphism φ_i on $\mathrm{Mod}^+(\Sigma_{g-1}, 2n)$ which is nonzero on h_i^+ and vanishes on all boundary-twist factors and on $h_j^\pm$ for $j \neq i$. When h_i^+ and h_i^- have no conjugate nonzero powers, we also choose $\varphi_i(h_i^-) = 0$. Otherwise, the preceding argument gives $\varphi_i(h_i^-) = \varphi_i(h_i^+)$. Additivity on commuting elements therefore gives $\varphi_i(f_i) \neq 0$ and $\varphi_i(f_j) = 0$ for $j \neq i$. After rescaling, we have

$$\varphi_i(f_j) = \delta_{ij} \qquad (3 \leq i, j \leq N).$$

Put $q_i = \varphi_i \circ \Lambda$. Then $[\delta q_i] \in EH_b^2(\mathrm{Mod}(N_g, n))$. Suppose

$$\sum_{i=3}^{N} a_i \iota_D^*[\delta q_i] = 0.$$

The homogeneous quasimorphism $\sum_{i=3}^N a_i(q_i \circ \iota_D)$ represents zero in exact bounded cohomology and hence is a homomorphism. By (8), it vanishes on every v_j. Evaluating gives

$$0 = \sum_{i=3}^{N} a_i q_i(\iota_D(v_j)) = a_j \qquad (3 \leq j \leq N).$$

The restricted classes are linearly independent. Since N is arbitrary, the image is infinite dimensional. In degree two there are no nontrivial bounded cohomology classes of zero seminorm; see [26, Corollary 2.7]. Hence the reduced assertion follows. □

A class detected in a disc. Choose an oriented disc $D_0 \subset N_g \setminus Q$ and a base configuration of three points in its interior. In this disc, let σ_1, σ_2 denote the standard Artin generators of $B_3(D_0)$. Let $\iota : B_3(D_0) \hookrightarrow B_3(N_g)$ be the local inclusion, and put

$$\jmath : F_2 \hookrightarrow B_3(D_0), \qquad \jmath(a) = \sigma_1^2, \quad \jmath(b) = \sigma_2^2.$$

The image of $\jmath$ lies in $P_3(D_0)$. The subgroup generated by σ_1^2, σ_2^2 is the free factor in $P_3(D_0) \cong F_2 \times \mathbb{Z}$.

Lemma 5. *There is a homogeneous quasimorphism $q_0 : B_3(N_g) \to \mathbb{R}$ such that*

$$\eta = \jmath^*\iota^*[\delta q_0] \neq 0 \quad \text{in } \overline{H}^2_b(F_2).$$

Proof. Apply Lemma 4 with $n = 3$ and $D = D_0$. It gives a homogeneous quasimorphism on $\mathrm{Mod}(N_g, 3)$ whose restriction to $B_3(D_0)$ is not a homomorphism. Pull it back by the point-pushing homomorphism $B_3(N_g) \to \mathrm{Mod}(N_g, 3)$, and denote the resulting homogeneous quasimorphism by q_0. The composite

$$B_3(D_0) \xrightarrow{\iota} B_3(N_g) \longrightarrow \mathrm{Mod}(N_g, 3)$$

is extension by the identity, so $\iota^*[\delta q_0] \neq 0$.

Restriction to the finite-index subgroup $P_3(D_0)$ is injective in bounded cohomology. The amenable direct factor in $P_3(D_0) \cong F_2 \times \mathbb{Z}$ can then be removed isometrically. Thus $\jmath^*\iota^*[\delta q_0] \neq 0$. In degree two there are no nontrivial bounded cohomology classes of zero seminorm; see [26, Corollary 2.7]. Hence this class remains nonzero after reduction. □

Remark. A quasimorphism satisfying Lemma 5 can also be constructed directly. We identify $B_3(N_g)$ with its point-pushing subgroup of $\mathrm{Mod}(N_g, 3)$ and use the lift Λ defined above. Interpret the word $v_3 = u_3^6\Delta_3^2$ from (7) in $B_3(D_0)$, using the standard Artin generators of this disc braid group. The projective matrix of u_3 is $[M_3] = \left[\begin{smallmatrix} 4 & 9 \\ 7 & 16 \end{smallmatrix}\right]$. No nonzero power of $[M_3]$ is conjugate to its inverse in $\mathrm{PGL}(2, \mathbb{Z})$; see [1, Example 2, Lemma 9 and Proposition 5].

Applying the argument in the proof of Lemma 4 to the single braid v_3, we obtain a homogeneous quasimorphism φ on $\mathrm{Mod}^+(\Sigma_{g-1}, 6)$ such that

$$\varphi\big(\Lambda(\iota(v_3))\big) \neq 0.$$

Let q_0 be the restriction of $\varphi \circ \Lambda$ to $B_3(N_g)$. Then $q_0(\iota(v_3)) \neq 0$. Since $v_3 \in [B_3(D_0), B_3(D_0)]$, the restriction of q_0 to $B_3(D_0)$ is not a homomorphism. Restriction to $P_3(D_0)$ and removal of its central $\mathbb{Z}$ factor, as in the proof of Lemma 5, give $\jmath^*\iota^*[\delta q_0] \neq 0$ in $\overline{H}^2_b(F_2)$.

The mixed product transfer. Let $\alpha \in \overline{H}^2_b(B_3(N_g))$ be the reduced class of δq_0, and put

$$A = \Gamma^2_{N_g,3}(\alpha) \in \overline{H}^2_b(\mathcal{T}_{N_g}).$$

Recall that $\eta = \jmath^*\iota^*\alpha \neq 0$ in $\overline{H}^2_b(F_2)$ by Lemma 5. For $m \geq 0$ and $q \in \{2, 3\}$, define

$$\begin{aligned} \Phi_{m,q} &: \overline{H}^q_b(F_2) \longrightarrow \overline{H}^{2m+q}_b(\mathcal{T}_{N_g}), \\ \Phi_{m,q}(v) &= A^{\smile m} \smile \Theta_q(v). \end{aligned} \tag{9}$$

Here $A^{\smile 0}$ is the degree-zero unit. This is the mixed transfer of $\alpha^{\times m} \times J_q(v)$, a class on $B_3(N_g)^m \times \pi_1(N_g)$, over $X_3(N_g)^m \times N_g$ with measure $\nu^m \times \mu$.

Indeed, the cup-product integrand factors over the independent variables, and Fubini gives (9).

For the testing argument fix $m \geq 1$. Choose pairwise disjoint discs $D_1, \dots, D_m \Subset \mathrm{int}(D_0)$. Choose $R > 0$ sufficiently small. Inside each D_j, choose two overlapping round discs $D_{j,1}, D_{j,2}$ of radius R in a positively oriented area coordinate chart, with closures in D_j, such that their intersection and both exclusive regions have positive measure. Label these three regions in the order $D_{j,1} \setminus D_{j,2}$, $D_{j,1} \cap D_{j,2}$, $D_{j,2} \setminus D_{j,1}$. Use a model configuration with one point in each region, in the displayed order.

For sufficiently small $0 < \varepsilon < R$, choose a smooth function $\chi_\varepsilon : [0, R] \to [0, 1]$ equal to one on $[0, R - \varepsilon]$ and zero near R. For $s = 1, 2$, use polar coordinates (r, φ) about the center of $D_{j,s}$, choosing the increasing angular direction so that the inverse traced full rotation of the model configuration represents σ_s^2. As in [5, Lemmas 15 and 21], let $f_{j,s,\varepsilon}$ be the time-one map of the rotation

$$(r, \varphi) \longmapsto (r, \varphi + 2\pi t \chi_\varepsilon(r)), \qquad 0 \leq t \leq 1,$$

on $D_{j,s}$, extended by the identity outside that disc. The transition collars are the regions where $R - \varepsilon < r < R$; their total measure tends to zero as $\varepsilon \to 0$. Define

$$\rho_{j,\varepsilon} : F_2 \longrightarrow \mathcal{T}_{N_g}, \qquad \rho_{j,\varepsilon}(a) = f_{j,1,\varepsilon}, \quad \rho_{j,\varepsilon}(b) = f_{j,2,\varepsilon}.$$

Fix j, ε, and let $\mathbf{x} \in X_3(N_g)$ have one point in each of the three regions and no point in either transition collar. Both generators fix $\mathbf{x}$. Hence,

$$F_2 \longrightarrow B_3(N_g), \qquad w \longmapsto \gamma_{N_g,3}(\rho_{j,\varepsilon}(w), \mathbf{x})$$

is a homomorphism. Order the points by their regions. Choose an orientation-preserving identification of the pair $D_{j,1}, D_{j,2}$, together with these points, with the standard pair of overlapping discs in D_0 surrounding base points $1, 2$ and $2, 3$, respectively. The inverse trajectory braids of the two rotations are then σ_1^2 and σ_2^2. Restoring the original labels and using the chosen closing paths conjugates both braids by the same element. Thus there is $\beta_{\mathbf{x}} \in B_3(N_g)$ such that

$$\gamma_{N_g,3}(\rho_{j,\varepsilon}(w), \mathbf{x}) = \beta_{\mathbf{x}} (\iota \circ \jmath)(w) \beta_{\mathbf{x}}^{-1} \qquad (w \in F_2).$$

For the fixed j, ε, the braid $\beta_{\mathbf{x}}$ may depend on $\mathbf{x}$ and the closing paths, but not on w; see [5, proofs of Lemmas 15 and 21]. Since inner automorphisms act trivially on bounded cohomology, the pullback of α by this homomorphism is $\jmath^* \iota^* \alpha = \eta \in \overline{H}_b^2(F_2)$.

The images for different j commute, and their supports are disjoint from the support of s_δ, which lies in Q. Thus

$$\begin{gathered} \rho_{\varepsilon,\delta} : F_2^{m+1} \longrightarrow \mathcal{T}_{N_g}, \\ \rho_{\varepsilon,\delta}(g_1, \dots, g_m, h) = \rho_{1,\varepsilon}(g_1) \cdots \rho_{m,\varepsilon}(g_m) s_\delta(h) \end{gathered} \tag{10}$$

is a homomorphism; compare [5, equation (63)]. The annular pushes s_δ and the disc rotations $f_{j,s,\varepsilon}$ are time-one maps of smooth ω_0-preserving isotopies; hence $\rho_{\varepsilon,\delta}$ takes values in $\mathrm{Diff}_0(N_g,\omega_0)$. Let $p_j : F_2^{m+1} \to F_2$ be the coordinate projections, and put

$$\eta_{j,\varepsilon} = \rho_{j,\varepsilon}^* A, \qquad E_\delta = s_\delta^* A, \qquad v_\delta = s_\delta^* \Theta_q(v).$$

Thus $\eta_{j,\varepsilon}, E_\delta \in \overline{H}_b^2(F_2)$ and $v_\delta \in \overline{H}_b^q(F_2)$. The degree-two splitting of [5, Lemma 22] gives (11), and [5, Lemma 24] gives (12):

$$\rho_{\varepsilon,\delta}^* A = \sum_{j=1}^{m} p_j^* \eta_{j,\varepsilon} + p_{m+1}^* E_\delta, \tag{11}$$

$$\rho_{\varepsilon,\delta}^* \Theta_q(v) = p_{m+1}^* v_\delta. \tag{12}$$

Let $\lambda_j = 6\,\mu(D_{j,1} \setminus D_{j,2})\,\mu(D_{j,1} \cap D_{j,2})\,\mu(D_{j,2} \setminus D_{j,1}) > 0$. By [5, Lemma 21] and (6),

$$\eta_{j,\varepsilon} \longrightarrow \lambda_j \eta \quad (\varepsilon \to 0), \qquad v_\delta \longrightarrow \kappa v \quad (\delta \to 0)$$

in reduced norm. The defining isotopies of $\rho_{j,\varepsilon}$ stay in the contractible disc D_j, and hence give trivial one-point trajectory classes. This is why the first m factors do not occur in (12).

Two coordinates. For $m = 1$, naturality of the cup product gives

$$\rho_{\varepsilon,\delta}^* \Phi_{1,q}(v) = \eta_{1,\varepsilon} \times v_\delta + p_2^*(E_\delta \smile v_\delta).$$

Let $\tau_1 : F_2^2 \to F_2^2$ be given by $\tau_1(g,h) = (1,h)$. Applying $\mathrm{id} - \tau_1^*$ removes the second summand. Hence

$$(\mathrm{id} - \tau_1^*)\rho_{\varepsilon,\delta}^* \Phi_{1,q}(v) \longrightarrow \lambda_1 \kappa\, \eta \times v \qquad ((\varepsilon,\delta) \to (0,0)).$$

If $\Phi_{1,q}(v) = 0$, then $\eta \times v = 0$. Since $v \mapsto \eta \times v$ is injective by Proposition 3, this gives $v = 0$.

An arbitrary number of disc factors. For $1 \le j \le m$, let $\tau_j : F_2^{m+1} \to F_2^{m+1}$ replace coordinate j by the identity, and put

$$\Pi_m = \prod_{j=1}^{m} (\mathrm{id} - \tau_j^*).$$

We follow the calculation in [5, Section 5.1, equations (68)–(70)]. Expand (9) using (11)–(12). There are exactly m degree-two factors to distribute. A term using E_δ, or assigning two factors to the same disc coordinate, leaves another disc coordinate unused and is sent to zero by Π_m. The remaining terms assign one factor to each disc coordinate. There are $m!$ such assignments, and their signs agree because the factors have even degree. Thus

$$\Pi_m \rho_{\varepsilon,\delta}^* \Phi_{m,q}(v) = m!\, \eta_{1,\varepsilon} \times \cdots \times \eta_{m,\varepsilon} \times v_\delta.$$

Passing to the limit in reduced norm as $(\varepsilon,\delta)\to(0,0)$ gives

$$\Pi_m\rho^*_{\varepsilon,\delta}\Phi_{m,q}(v)\longrightarrow m!\kappa\Big(\prod_{j=1}^{m}\lambda_j\Big)\eta^{\times m}\times v. \tag{13}$$

If $\Phi_{m,q}(v)=0$, the left-hand side of (13) is zero for every ε,δ. Thus $\eta^{\times m}\times v=0$ in reduced bounded cohomology. Proposition 3 gives $v=0$. For $m=0$, injectivity is (6).

Exactness. By (5) and the one-point comparison identity, $\Theta_q(v)$ lies in $\overline{EH}^q_b(\mathcal{T}_{N_g})$. Cup products with exact classes are exact before reduction, since comparison respects cup products. Passing to the reduced groups shows that $\Phi_{m,q}(v)\in\overline{EH}^{2m+q}_b(\mathcal{T}_{N_g})$. Thus we have injective maps

$$\Phi_{m,q}:\overline{H}^q_b(F_2)\longrightarrow\overline{EH}^{2m+q}_b(\mathcal{T}_{N_g}),\qquad m\geq 0,\quad q=2,3.$$

Every integer $d\geq 2$ has the form $d=2m+q$. The same testing homomorphisms prove injectivity after restricting $\Phi_{m,q}$ to $\mathrm{Diff}_0(N_g,\omega_0)$, and exactness is preserved under restriction. Thus Equation (1) proves Theorem 1 in dimension two for the fixed measure μ and density ω_0. Lemma 2 and Moser's theorem give the assertions for every good measure and every smooth positive density, respectively.

4. Proof of Theorem 1 in dimension greater than two

The free subgroup. Let $\dim M=n\geq 3$. Retain $G=\pi_M=\pi_1(M)$ and $\mathcal{T}_M=\mathrm{Homeo}_0(M,\mu)$.

Lemma 6. *There is a hyperbolic embedding $i:F_2\hookrightarrow G$ induced by an embedded figure-eight in M with an orientable regular neighborhood. For every integer $k\geq 1$, the same embedding can be realized, after choosing basepoint paths, by k disjoint embedded figure-eights with disjoint orientable regular neighborhoods.*

Proof. Let $w_M:G\to\mathbb{Z}/2$ be the orientation character: it is zero on loops preserving local orientation and one on loops reversing it. The group G is torsion-free and nonelementary hyperbolic, and $\ker w_M$ has index two. By [22, Theorem 7.7], $\ker w_M$ contains a free subgroup of rank two which is malnormal and quasiconvex in the full group G. Such a subgroup is hyperbolically embedded; see [4, Theorem 7.11] and [14, Proposition 4.28(a)]. Denote the resulting marked inclusion by $i:F_2=\langle a,b\rangle\hookrightarrow G$.

Represent $i(a),i(b)$ by a map of a figure-eight. General position makes this map an embedding without changing the induced inclusion on fundamental groups. Take a regular neighborhood U. Since $i(F_2)\subset\ker w_M$, every loop in U preserves local orientation in M. Thus U is orientable.

Apply general position simultaneously to k copies of the marked graph map inside U. The resulting figure-eights are pairwise disjoint, and have disjoint small regular neighborhoods. □

The local maps and Moser's theorem. Fix $k \geq 1$ and choose an orientation on each neighborhood in Lemma 6. There the density defining μ is represented by a positive volume form ω. Write D^{n-1}_r for the closed Euclidean ball of radius r, and put $D^{n-1} = D^{n-1}_1$. In one neighborhood, the two loops have trivial normal bundles. Choose tubes $T_a, T_b \cong S^1 \times D^{n-1}$ around them whose intersection is a ball. On the model tube, use coordinates $\vartheta \in \mathbb{R}/\mathbb{Z}$ and $y = (y_1, \ldots, y_{n-1}) \in D^{n-1}$, and put

$$Q_n = (\mathbb{R}/\mathbb{Z}) \times D^{n-1}, \qquad \omega_0 = \mathrm{d}\vartheta \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_{n-1}.$$

For $s = a, b$, put $c_s = \mu(T_s)/\int_{Q_n} \omega_0 > 0$. Moser's theorem in the boundary version of Banyaga [27, 2], applied to these forms of equal total volume, gives a parametrization

$$\psi_s : T_s \longrightarrow Q_n, \qquad \psi_s^*(c_s\omega_0) = \omega|_{T_s}. \tag{14}$$

Choose the angular directions so that the trajectory cocycle of one full circuit is $i(a)$, respectively $i(b)$, with the convention of Section 2. Thus the geometric circuits represent $i(a)^{-1}, i(b)^{-1}$. For $0 < \varepsilon < 1$, let $\chi_\varepsilon : D^{n-1} \to [0,1]$ be a smooth function equal to one on $D^{n-1}_{1-\varepsilon}$ and zero near the boundary. The isotopy

$$P^t_\varepsilon(\vartheta, y) = (\vartheta + t\chi_\varepsilon(y), y), \qquad 0 \leq t \leq 1 \tag{15}$$

preserves ω_0. Conjugate by ψ_s and extend by the identity outside T_s. The time-one map fixes the inner region $\psi_s^{-1}((\mathbb{R}/\mathbb{Z}) \times D^{n-1}_{1-\varepsilon})$ pointwise, while its trajectories there make one full circuit. The complementary collar in T_s has measure tending to zero as $\varepsilon \to 0$.

This is the construction of Brandenbursky–Menashe [9, Section 4.5 and Lemma 4.12] in dimension n; see also the tube formula in [8, Lemma 3.1]. Here is where we use the orientability of the figure-eight neighborhood: it gives the positive volume forms to which Moser's theorem applies in (14). Apply this construction in k disjoint neighborhoods $U_1, \ldots, U_k$. Write $T_{j,a}, T_{j,b} \subset U_j$ for the two tubes and $\psi_{j,s}$ for their parametrizations, where $s = a, b$. Define

$$f_{j,s,\varepsilon}(x) = \begin{cases} \psi_{j,s}^{-1} P^1_\varepsilon \psi_{j,s}(x), & x \in T_{j,s}, \\ x, & x \notin T_{j,s}. \end{cases}$$

Let a_j, b_j be the standard generators of the jth factor of F_2^k. We define

$$\rho_\varepsilon : F_2^k \longrightarrow \mathcal{T}_M, \qquad \rho_\varepsilon(a_j) = f_{j,a,\varepsilon}, \quad \rho_\varepsilon(b_j) = f_{j,b,\varepsilon}. \tag{16}$$

For $w_1, \ldots, w_k \in F_2$, define

$$\rho_\varepsilon(w_1, \ldots, w_k) = \prod_{j=1}^{k} w_j(f_{j,a,\varepsilon}, f_{j,b,\varepsilon}),$$

where each word is evaluated at the two indicated maps. Since maps corresponding to distinct indices j have disjoint supports and therefore commute, ρ_ε is a homomorphism. Each $f_{j,s,\varepsilon}$ arises from a smooth ω_0-preserving isotopy, so ρ_ε actually takes values in $\mathrm{Diff}_0(M,\omega_0)$. For each tube define its inner region

$$A_{j,s,\varepsilon} = \psi_{j,s}^{-1}\big((\mathbb{R}/\mathbb{Z}) \times D^{n-1}_{1-\varepsilon}\big), \qquad s = a, b.$$

Let B_ε be the union of all collars, and let $U_{j,\varepsilon}$ be the overlap of the two inner regions in U_j:

$$B_\varepsilon = \bigcup_{j=1}^{k} \bigcup_{s=a,b} (T_{j,s} \setminus A_{j,s,\varepsilon}), \qquad U_{j,\varepsilon} = A_{j,a,\varepsilon} \cap A_{j,b,\varepsilon}.$$

Then $\mu(B_\varepsilon) \to 0$, and every element of $\rho_\varepsilon(F_2^k)$ fixes $M \setminus B_\varepsilon$ pointwise. In particular, B_ε is invariant under $\rho_\varepsilon(F_2^k)$. Write $p_j : F_2^k \to F_2$ for the coordinate projection. For $x \in U_{j,\varepsilon}$, the trajectory homomorphism $w \mapsto \gamma(\rho_\varepsilon(w), x)$ is conjugate to $i \circ p_j$. On the remaining fixed regions its image is cyclic or trivial; see [5, Lemma 11]. Moreover,

$$\lambda_{j,\varepsilon} = \mu(U_{j,\varepsilon}) \longrightarrow \lambda_j = \mu(T_{j,a} \cap T_{j,b}) > 0.$$

Transfer and injectivity. For the inclusion i above, the extension theorem of Frigerio–Pozzetti–Sisto [20, Corollary 2 and Proposition 5.1] gives, for $q = 2, 3$, algebraic linear sections

$$J_q : \overline{H}{}^q_b(F_2) \longrightarrow \overline{EH}{}^q_b(G), \qquad i^* J_q = \mathrm{id};$$

see [5, Lemma 6]. Choose the nonzero class $\theta \in \overline{H}{}^2_b(F_2)$ supplied by [5, Lemma 7]. It satisfies

$$\theta \smile u = 0 \qquad (u \in \overline{H}{}^s_b(F_2),\ s \geq 1).$$

In this section, T^q_m denotes the map of Proposition 3 for this choice of θ. For $m \geq 0$ and $q = 2, 3$, put $d = 2m + q$, $k = m + 1$, and define

$$\Phi_{m,q} : \overline{H}{}^q_b(F_2) \longrightarrow \overline{EH}{}^d_b(\mathcal{T}_M), \qquad \Phi_{m,q}(v) = \overline{\Gamma}{}^d_{b,k}\big(J_2(\theta)^{\times m} \times J_q(v)\big). \tag{17}$$

The source factors have representatives in exact bounded cohomology. External products commute with comparison, and the product transfer preserves exactness. Hence $\Phi_{m,q}(v) \in \overline{EH}{}^d_b(\mathcal{T}_M)$.

Permutation invariance of the product measure [5, equation (26)] gives

$$\Phi_{m,q}(v) = \frac{1}{k!}\, \overline{\Gamma}{}^d_{b,k}\big(\mathrm{Sym}_k(J_2(\theta)^{\times m} \times J_q(v))\big).$$

Put $C_k = \prod_{j=1}^k \lambda_j > 0$ and apply [5, Proposition 12] to ρ_ε. The proof partitions M^k according to the trajectory types just described. The set where an integration coordinate lies in B_ε has measure at most $k\mu(B_\varepsilon)$, so its contribution tends to zero in norm. On its complement, the point-dependent conjugations do not affect the resulting bounded cohomology class; see [5, proof of Proposition 12]. A term with a cyclic or trivial trajectory factor vanishes in positive degrees. If two integration coordinates use the same free factor, restriction contains either $\theta \smile \theta$ or $\theta \smile v$, and hence vanishes. The surviving regions are the $k!$ products obtained by permuting $U_{1,\varepsilon}, \dots, U_{k,\varepsilon}$,

each with measure $\prod_j \lambda_{j,\varepsilon}$. Their sum gives $T_m^q(v)$, with the normalization above. Consequently,

$$\rho_\varepsilon^* \Phi_{m,q}(v) \longrightarrow C_k T_m^q(v) \qquad (\varepsilon \to 0) \tag{18}$$

in reduced norm for every fixed $v \in \overline{H}_b^q(F_2)$.

If $\Phi_{m,q}(v) = 0$, then (18) and $C_k > 0$ give $T_m^q(v) = 0$. Proposition 3 implies $v = 0$, so $\Phi_{m,q}$ is injective. Since the testing maps ρ_ε take values in $\mathrm{Diff}_0(M, \omega_0)$, the same argument proves injectivity after restriction to this group; the restricted classes remain exact by naturality of comparison. By (1),

$$\dim \overline{EH}_b^{2m+q}(\mathcal{T}_M) = \infty \qquad (m \geq 0,\ q = 2, 3).$$

Every integer $d \geq 2$ has this form. Lemma 2 and the density version of Moser's theorem give the assertions for arbitrary good measures and smooth positive densities, respectively. Finally, reduced exact bounded cohomology is a quotient of unreduced exact bounded cohomology, so the unreduced assertions follow. This completes the proof of Theorem 1.

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